SKJ SOLUTIONS

ICT NETWORK

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ECTE 364 - DATA COMMUNICATIONS

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1. Executive Summary

Edu-Group Schools, a leading multi-campus K–12 educational institution in the UAE, faces significant challenges in managing its growing digital infrastructure. With three campuses serving over 15,000 students and 900 staff members, the existing ICT environment suffers from fragmented systems, inconsistent network performance, limited visibility into operations, and inadequate security controls. These limitations hinder teaching effectiveness, administrative efficiency, and the institution's ability to support modern digital learning initiatives.

SKJ Solutions proposes a comprehensive ICT modernization strategy that transforms Edu-Group Schools' digital infrastructure into a secure, intelligent, and future-ready ecosystem. The solution integrates an SD-WAN-enabled multi-campus network fabric, a Digital Twin-powered smart management dashboard, enterprise-grade cybersecurity architecture, and a unified communication and collaboration platform. This holistic approach provides centralized control, real-time visibility, and predictive intelligence across all campuses, ensuring reliable performance for academic and administrative operations.

The proposed solution delivers substantial benefits, including enhanced network reliability with reduced downtime, improved security posture through zero-trust segmentation and multi-factor authentication, predictive maintenance capabilities that anticipate failures before they occur, and seamless collaboration tools that connect students, teachers, and parents. Key technologies include Aruba Wi-Fi 6 access points, Fortinet next-generation firewalls, Azure cloud integration, and SKJ Solutions' proprietary Digital Twin platform with AI-driven analytics.

By implementing this solution, Edu-Group Schools will achieve operational excellence, future-proof its infrastructure, and create an intelligent learning environment that supports pedagogical innovation while ensuring data security and regulatory compliance. The transformation positions the institution as a leader in digital educational infrastructure, capable of adapting to technological advancements and scaling effectively with future growth.

2. Introduction

In today’s rapidly evolving educational landscape, robust and intelligent network infrastructure has become essential for delivering high-quality learning experiences and streamlined school operations. With digital learning, hybrid classrooms, and connected devices becoming central to modern education, institutions now need network systems that are not only fast and secure but also capable of adapting to continuous technological change. In this context, Edu-Group Schools is preparing to elevate its ICT environment to match the pace of innovation shaping the future of teaching and learning.

As one of the largest K–12 education providers in the UAE, Edu-Group Schools spans three campuses and serves a diverse community of more than 15,000 students and nearly 900 staff members. The institution has steadily adopted various digital tools, communication platforms, and smart learning technologies. However, the growing scale of operations, the rise of cloud-based services, and the expanding number of connected devices across campuses have created a strong need for a more cohesive, secure, and intelligently managed ICT ecosystem. The leadership aims to adopt solutions that not only integrate existing systems but also bring clarity, consistency, and long-term readiness to their digital infrastructure.

To support this vision, SKJ Solutions has been commissioned to design and implement a comprehensive ICT modernization strategy tailored to Edu-Group Schools. The proposed solution integrates SD-WAN-enabled intelligent networking, enterprise-grade cybersecurity, Digital Twin-powered monitoring for real-time visibility, and unified communication services for students, teachers, and parents. By bringing together advanced technologies with practical operational needs, the project aims to build a resilient and future-ready ICT environment that strengthens learning outcomes, enhances collaboration, and ensures reliable performance across all campuses. This report presents the strategic approach, technical architecture, implementation plan, and governance framework that will guide the transformation of Edu-Group Schools into a leader in digital educational infrastructure.

3. Company Background

3.1. Brief Introduction and Background

SKJ Solutions is a pioneering systems integration and intelligent network solutions company, founded in 2018 by a group of visionary ICT engineers and network architects. The company was established to address the growing demand for intelligent, adaptive, and digitally replicable network infrastructures in educational institutions and enterprises. From the outset, SKJ Solutions focused on creating high-performance, secure, and future-ready networks that go beyond traditional connectivity, integrating Digital Twin-enabled network management, IoT-driven monitoring, and data-driven operational systems.

Over the years, SKJ Solutions has developed a reputation for delivering innovative, reliable, and tailored network solutions, combining technical expertise with a deep understanding of client requirements. The company's approach emphasizes not only deployment and functionality but also long-term maintainability, scalability, and operational intelligence, ensuring clients gain maximum value from their ICT investments.

3.2. History of the Company

SKJ Solutions began with the design and deployment of small-scale campus networks, providing reliable connectivity to classrooms, labs, and administrative offices. Recognizing the potential of emerging technologies, the company quickly expanded its capabilities to include Digital Twin dashboards, predictive network monitoring, IoT-integrated classrooms, and advanced cybersecurity frameworks. Over time, SKJ Solutions has delivered a wide range of ICT projects across educational institutions and enterprises, establishing itself as a trusted partner for intelligent network solutions.

3.2.1. Previous Initiatives and Projects

Over the past several years, SKJ Solutions has led transformative ICT modernization programs across schools and educational institutions, building intelligent and resilient digital environments that elevate both teaching and operational performance. Each project reflects a deep commitment to crafting secure, future-ready ecosystems where infrastructure, learning platforms, and administrative systems work together seamlessly. Through these initiatives, SKJ Solutions has helped institutions strengthen digital reliability, enrich classroom experiences, and create technology foundations that continue to grow with their communities.

1. ***Smart Campus Network Transformation – Horizon International School (2021–2022)***

SKJ Solutions executed a full-scale modernization of Horizon International School's network backbone, replacing fragmented legacy systems with a unified, high-performance architecture. The project introduced Wi-Fi 6 across academic and administrative blocks, centralized switching and routing, and a role-based authentication model for staff and student devices. A campus-level Digital Twin was developed to map devices, simulate load conditions, and predict points of failure before they occurred. This allowed administrators to visualize network health in real time and test configuration changes safely, resulting in a significant reduction in downtime and improved classroom connectivity.

2. ***Multi-Campus SD-WAN and Hybrid Cloud Modernization – FutureMinds Education Group (2022–2023)***

For FutureMinds Education Group, SKJ Solutions delivered a highly scalable SD-WAN fabric connecting five geographically separated campuses under a unified policy framework. The implementation standardized routing, improved bandwidth utilization, and introduced automated failover for uninterrupted academic operations.

The project also transitioned critical services to a hybrid-cloud environment, hosting LMS workloads, ERP systems, and file services on a mix of on-premises and cloud platforms. Centralized dashboards were established to monitor application performance, network security, and user activity across all sites, giving the institution a consolidated view of its entire ICT estate.

3. ***AI-Driven IoT Monitoring and Digital Twin Deployment – Crescent Academy (2023–2024)***

At Crescent Academy, SKJ Solutions introduced an advanced IoT infrastructure integrating smart classroom devices, environmental sensors, and energy management systems into a single monitoring ecosystem. An AI-enabled Digital Twin was built to interpret sensor data, detect anomalies, and forecast equipment health trends. This allowed administrators to anticipate hardware failures, optimize energy consumption, and gain deeper insights into classroom utilization patterns. The Digital Twin served not only as a monitoring tool but also as a planning instrument, enabling the academy to model future expansions and adjust infrastructure capacity with precision.

3.3. Mission & Vision

Mission: SKJ Solutions operates with a mission to redefine digital connectivity by delivering intelligent, secure, and fully integrated network solutions that empower organizations with complete visibility, control, and operational insight across their ICT ecosystems.

Vision: The company envisions creating resilient, intelligent, and future-ready networks that are agile, adaptable, and capable of supporting rapid technological advancements. SKJ Solutions aims to enable clients to operate efficiently while maintaining stability, long-term maintainability, and operational continuity, ensuring their networks remain reliable and manageable for years to come.

Aims: SKJ Solutions is committed to providing its clients with robust, secure, and adaptive network infrastructures that not only meet current operational requirements but also anticipate future needs. The company seeks to integrate Digital Twin technologies and IoT solutions for real-time monitoring, predictive maintenance, and intelligent resource management. By offering unified communication platforms, advanced cybersecurity measures, and analytics-driven network management, SKJ Solutions ensures that organizations can operate efficiently and confidently in increasingly complex digital environments.

Outcomes: Clients partnering with SKJ Solutions can expect a comprehensive transformation of their ICT landscape. Networks become fully visible, intelligently managed, and highly efficient, reducing downtime and enhancing productivity. Educational institutions benefit from smart, sensor-enabled classrooms, improving learning experiences while optimizing energy usage. Enterprises gain resilient, secure, and adaptable networks that accommodate evolving technological demands. Overall, SKJ Solutions's solutions enable organizations to achieve enhanced operational efficiency, future-proof infrastructure, and data-driven decision-making, ensuring sustainable growth and long-term success.

4. ICT Solution

4.1. Network Architecture & Design

The proposed ICT framework is fully customized to meet the operational, academic, and administrative needs of Edu-Group Schools. Rather than adopting a generic network design, the solution integrates multiple technologies into a unified ecosystem tailored specifically for a multi-campus educational environment.

The goal of the network is to design a robust, scalable, and secure network architecture for the school that supports:

- High-speed data access across classrooms, labs, and administrative offices.
- VLAN Segmentation for security (students, staff, IoT, guests).
- Support for Unified Communication & Collaboration systems.
- Real-time monitoring and integration with the Digital Twin dashboard.
- Efficient bandwidth allocation and QoS for critical services.

The customized solution includes:

- **SD-WAN Powered Multi-Campus Fabric:** A secure, resilient network that connects all three campuses with centralized control, intelligent routing, and encrypted communication.
- **Digital Twin-Enabled Monitoring:** A real-time virtual replica of the ICT infrastructure that visualizes device health, predicts failures, and identifies performance bottlenecks proactively.
- **Hybrid Hosting Model:** A balanced deployment of on-premises infrastructure and cloud-hosted services to ensure low latency, high availability, and scalable growth.
- **Multi-Layered Cybersecurity Architecture:** End-to-end protection through firewalls, NAC, IAM, MFA, and threat detection systems following a defense-in-depth approach.
- **Unified Communication and Collaboration Platform:** Integrated email, messaging, conferencing, LMS tools, and cloud storage that streamline communication between students, teachers, and administrators.
- **Scalable Hardware and Software Stack:** Enterprise-grade switches, servers, Wi-Fi 6 access points, and licensed software solutions that support future expansion without structural redesign.

This unified and custom-built ecosystem ensures fast, secure, and intelligent connectivity across all campuses, enabling the institution to operate efficiently while supporting advanced learning technologies and long-term digital growth.

4.1.1. Network Design

The proposed network architecture connects three branch campuses to a centralized Headquarters (HQ) hosted virtually in the cloud. To ensure high performance, secure communication, and simplified management across all locations, the solution uses SD-WAN (Software-Defined Wide Area Networking) as the primary inter-campus connectivity method. SD-WAN provides intelligent routing, encrypted tunnels, centralized control, and automatic failover, making it ideal for a multi-campus educational environment.

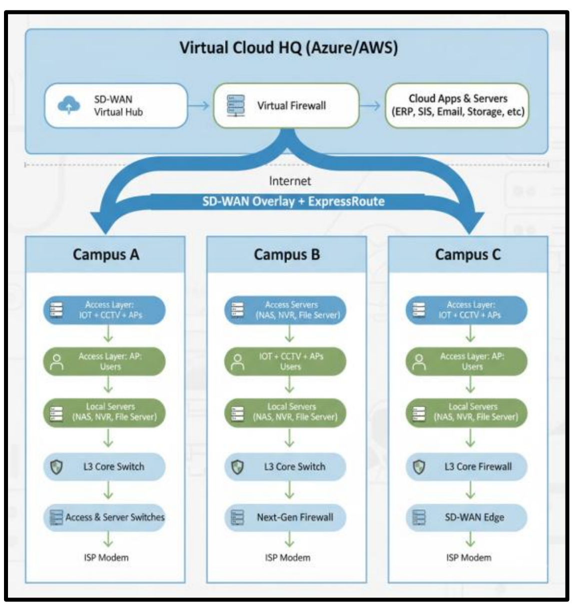

Fig 1: Network Design Topology

Each campus consists of a structured Local Area Network (LAN) that includes core switching, access layer switching, security firewalls, wireless infrastructure, servers, and end-user devices. All LAN components are integrated with the SD-WAN edge device, which securely connects each site to the virtual HQ through the SD-WAN overlay.

Each branch campus follows a standardized LAN design built around an ISP modem, a next-generation firewall for security, and an SD-WAN edge device that connects the site to the cloud HQ. The internal LAN uses a Layer 3 core switch that handles VLAN routing and segmentation, distributing connectivity to multiple access switches for end-user devices. These access switches support Wi-Fi 6 access points, IP cameras, CCTV NVRs, biometric systems, printers, smartboards, and student/admin PCs. A separate server switch hosts local resources such as file servers, NAS backups, or application servers, ensuring essential services remain available even during WAN outages.

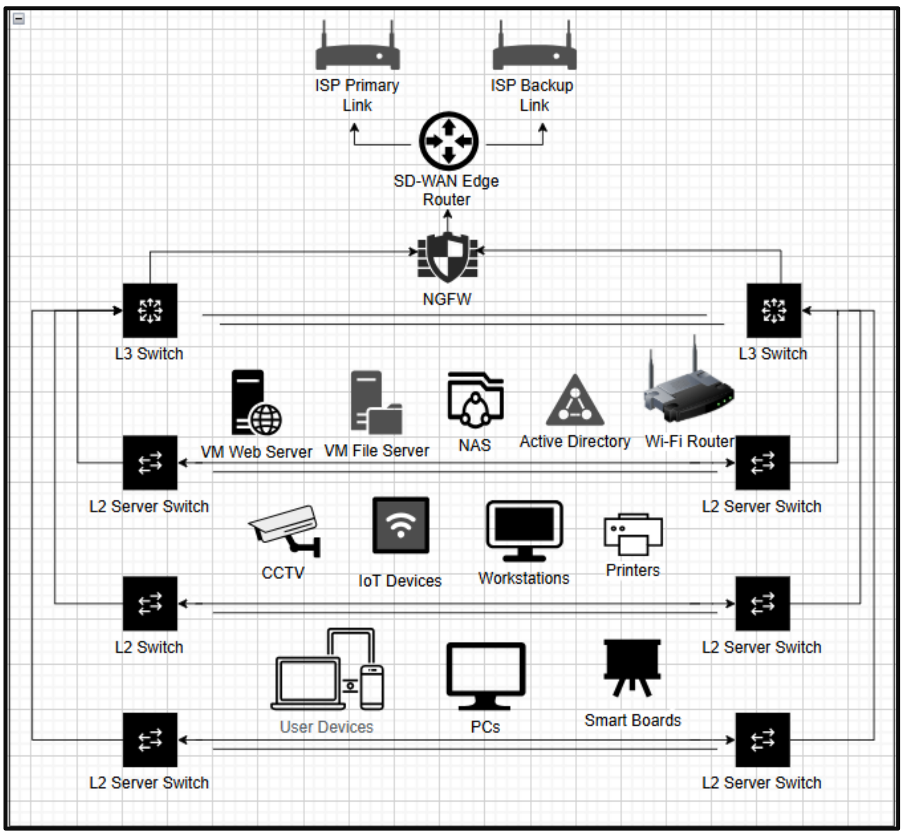

Fig. 2: Campus LAN

The SD-WAN edge device at each branch establishes encrypted tunnels to a virtual SD-WAN hub hosted in the cloud. This virtual HQ includes a virtual firewall, routing functions, and cloud-hosted application servers such as ERP, email, and school management platforms. SD-WAN enables centralized control, automated routing, dynamic path selection, and bandwidth optimization. Using cloud-hosted infrastructure allows scalable performance, easier maintenance, and simplifies inter-campus connectivity.

SD-WAN creates secure, high-performance overlay links allowing direct communication between campuses without routing traffic through a physical HQ. This improves latency for collaboration tools, VoIP, and real-time applications. It also provides automatic failover, intelligent path steering, and continuous monitoring across all campuses. With end-to-end encryption and centralized analytics, the SD-WAN overlay ensures reliable, secure, and efficient connectivity for all three branches and the virtual HQ.

The network deploys a hybrid hosting model, where each branch campus leverages a combination of local virtual infrastructure and cloud-hosted services. Critical applications, such as file servers, local backups, or latency-sensitive systems, are hosted on on-premises virtual machines at each site to ensure availability during WAN outages. The remaining services, including ERP, email, and school management platforms, are hosted in the cloud for scalability, simplified maintenance, and centralized management. This approach balances performance, redundancy, and cost-efficiency, providing campuses with reliable access to essential resources while benefiting from the flexibility and scalability of cloud infrastructure.

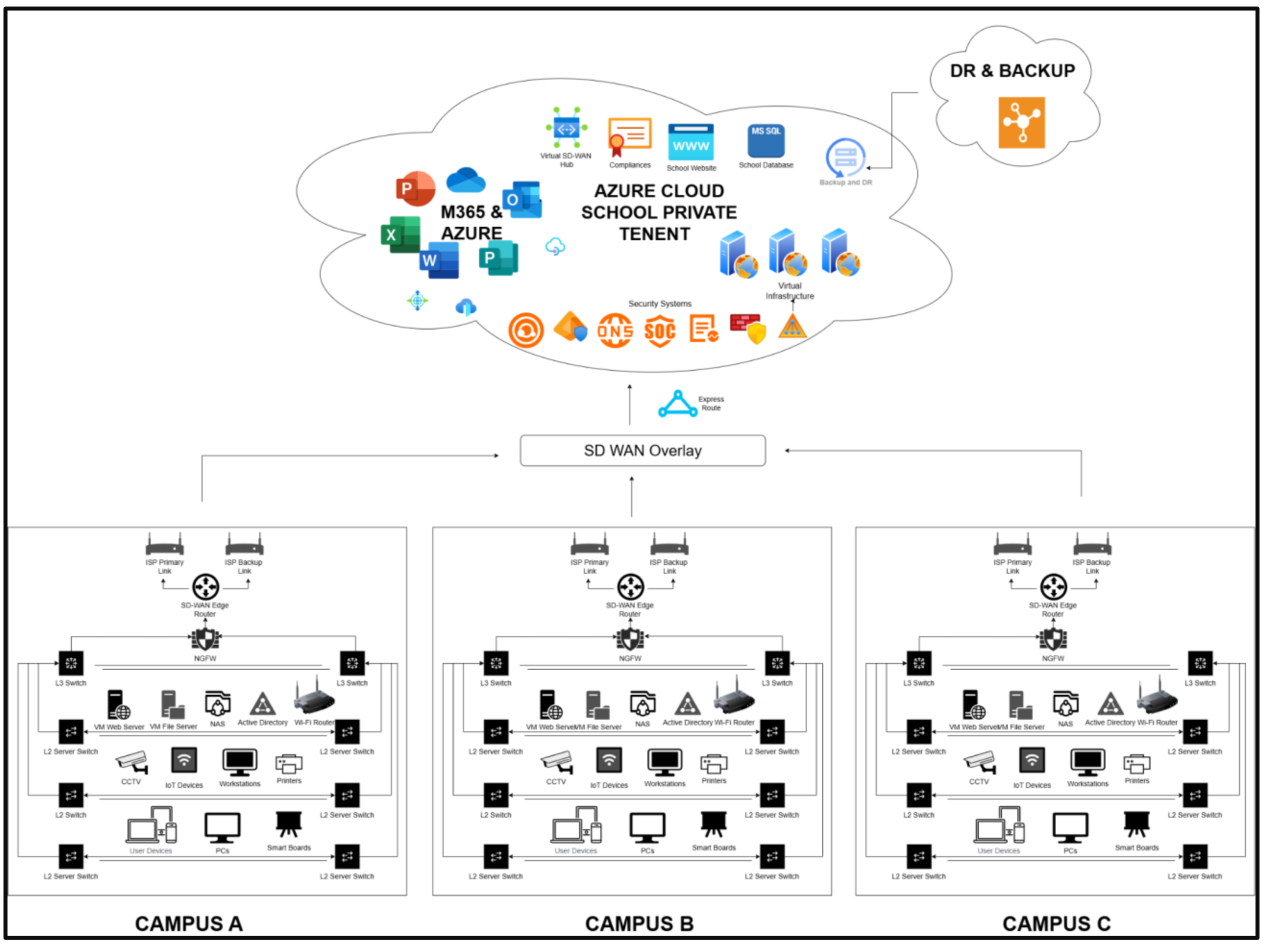

Fig. 3: Campus Network Architecture

4.1.2. Network Segmentation

The network employs both physical and logical segmentation to improve security, manageability, and performance. At the physical level, the network implements physical segmentation to separate different types of devices and traffic for improved security and performance. Dedicated switches are deployed for servers, access points, IoT devices, and end-user devices, ensuring that critical systems like local servers and CCTV NVRs are isolated from general user traffic. This separation reduces congestion, minimizes the risk of unauthorized access, and allows easier management and troubleshooting of each device category within the campus LAN.

At the logical level, VLANs are used to segment the school's LAN into separate virtual networks, improving performance, security, and traffic management across all three campuses. Each VLAN is assigned to a specific group of users or devices based on their operational role within the school. This segmentation approach reduces broadcast traffic, improves network performance, and enforces strict security boundaries between different departments.

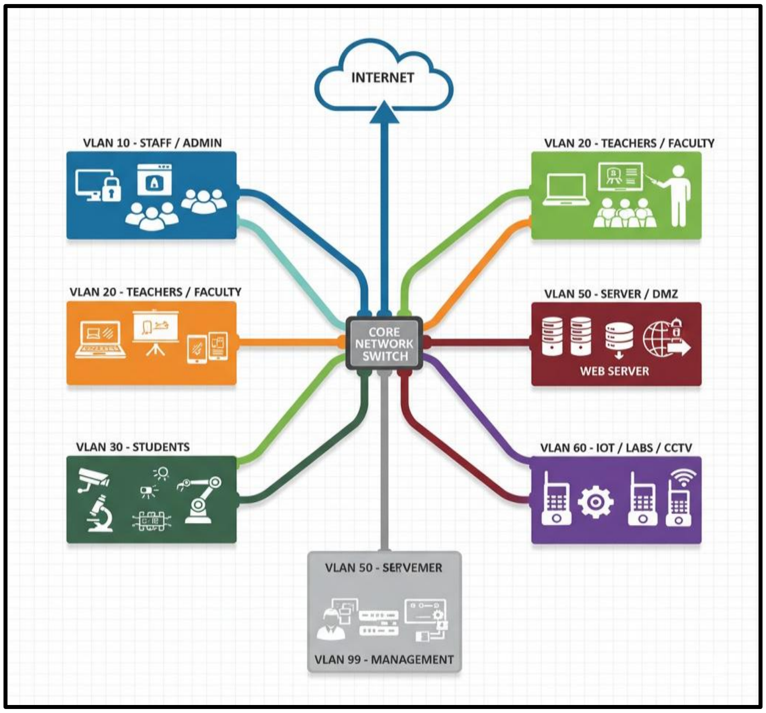

Fig. 4: VLAN Segmentation

The implemented VLAN scheme is as follows:

- **VLAN 10 - Staff / Admin:** Used for administrative computers, ERP access, and all internal staff applications.
- **VLAN 20 - Teachers / Faculty:** Dedicated to faculty devices, smart boards, and teaching-related applications.
- **VLAN 30 - Students:** Designed for student laptops, tablets, and lab computers.
- **VLAN 40 - Guest Wi-Fi:** Provides internet-only access to visitors.
- **VLAN 50 - Server / DMZ:** Hosts internal servers, ERP systems, school management applications, and any externally accessible services.

- **VLAN 60 - IoT / Labs / CCTV:** Contains IP cameras, lab equipment, IoT sensors, and other non-user devices.
- **VLAN 70 - Voice (IP Phones):** Separates VoIP systems from data traffic to ensure call quality, prioritize voice packets, and simplify QoS configuration.
- **VLAN 99 - Management:** Reserved for switch management, AP controllers, and network administrators.

This VLAN segmentation is implemented through Access Control Lists (ACLs). ACLs are applied on both switches and firewalls to enforce strict communication rules between VLANs and protect sensitive resources. On the Layer 3 core switch, ACLs limit unnecessary inter-VLAN traffic, such as preventing student VLANs from accessing server or management VLANs, while still allowing essential services like DNS or authentication. At the firewall, more advanced ACLs and security policies inspect and filter traffic between internal networks and the cloud, ensuring that only authorized devices and applications can reach critical systems such as ERP servers, the virtual HQ, or administrative services. This layered approach ensures that segmentation is not only logical but also securely enforced at every point in the network.

4.1.3. Hybrid Hosting

The proposed ICT solution adopts a hybrid hosting model that integrates both on-premises infrastructure at each campus and cloud-hosted services in a centralized virtual environment. This approach ensures high availability, low operational latency, and efficient delivery of critical academic and administrative applications across all three campuses.

The hybrid architecture brings significant operational and technical advantages:

- High Availability: Critical local services remain online even during WAN disruptions.
- Scalability: Cloud-hosted applications expand easily as the school grows.
- Lower Latency: Local hosting reduces delays for authentication, file access, and classroom tools.
- Centralized Management: Cloud-based oversight simplifies monitoring, updates, and analytics.
- Cost Efficiency: Reduces dependence on physical data centers and minimizes hardware load at each campus.
- Security: Segregated on-prem systems combined with cloud IAM, MFA, and access policies provide end-to-end protection.

On-Premises Components

Each campus maintains a dedicated set of core systems to ensure continuous operation, even during WAN or cloud outages. These include:

- Core Layer 3 Switches for local routing between VLANs
- Next-Generation Firewalls for perimeter protection and segmentation
- SD-WAN Edge Devices providing secure inter-campus connectivity
- Local Virtual Machines & Servers hosting AD, file servers, and essential applications
- Network Attached Storage (NAS/SAN) for local backups and high-speed storage
- Wireless LAN infrastructure including Wi-Fi 6 access points and controllers

These components handle authentication, internal services, and time-sensitive workloads while maintaining operational continuity during internet disruptions.

Cloud Infrastructure Components
The cloud environment hosts all centralized applications and scalable services, including:

- Virtual machines for ERP, LMS, email, and web hosting
- Cloud databases for structured data and system records
- Backup and disaster recovery instances
- Role-based identity and authentication services (Azure AD)
- API gateways enabling secure campus-to-cloud integration

Cloud deployment allows for elastic scaling as the number of students, devices, and services grows.

SD-WAN Connectivity Layer
The SD-WAN fabric forms the communication bridge between on-premises systems and cloud services. It provides:

- Encrypted tunnels (IPsec) and secure overlay networks
- Automated traffic steering based on application performance
- WAN failover and redundancy to minimize downtime
- Centralized control for all inter-campus and cloud routes

This layer ensures seamless communication between the three campuses and the virtual HQ.
Together, this hybrid mode is designed to balance performance, resilience, and scalability, allowing latency-sensitive services to run locally while leveraging the cloud for centralized applications, storage, and management.

4.1.4. Quality of Service (QoS)

Quality of Service (QoS) is implemented to ensure reliable performance for critical applications such as VoIP, video conferencing, digital classrooms, and cloud-hosted services. QoS identifies and prioritizes essential traffic to maintain low latency, minimal packet loss, and consistent bandwidth availability across all three campuses.

The QoS framework includes:

- **Traffic Classification:** Network traffic is categorized into voice, video, data, IoT, and best-effort classes using application signatures and DSCP markings.
- **Prioritization:** High-priority applications such as VoIP and live teaching sessions receive expedited forwarding treatment to guarantee consistent performance.
- **Bandwidth Management:** Minimum bandwidth thresholds are reserved for mission-critical services, while non-critical traffic is rate-limited during congestion.
- **Congestion Control:** Traffic shaping, queuing algorithms, and load management ensure stable performance during peak usage hours.
- **Continuous Monitoring:** QoS metrics are monitored through the Digital Twin, SD-WAN layer and network analytics tools in the cloud, enabling automatic adjustments and long-term optimization.

This QoS implementation ensures that academic applications, administrative systems, and real-time communication platforms remain uninterrupted, even under heavy network load.

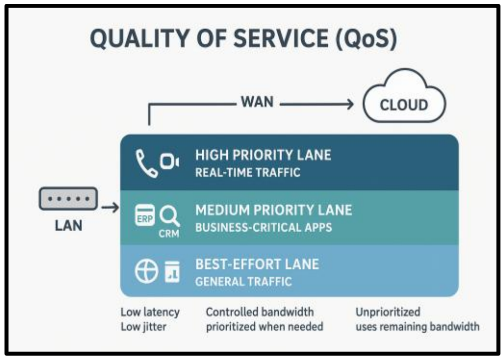

Fig. 5: Quality of Service

4.2. Digital Twin & Smart Management Dashboard

4.2.1. Overview

The proposed solution introduces a digital twin platform that creates a virtual representation of the Edu-Group Schools' entire network infrastructure. This digital environment mirrors the physical layout of all three campuses, the devices operating within them, and the connections linking them together. By integrating real-time monitoring, topology awareness, and predictive insights, the digital twin serves as a single source of truth for administrators. It allows the IT team to understand the current state of the network, diagnose issues rapidly, and anticipate failures before they affect teaching and learning operations.

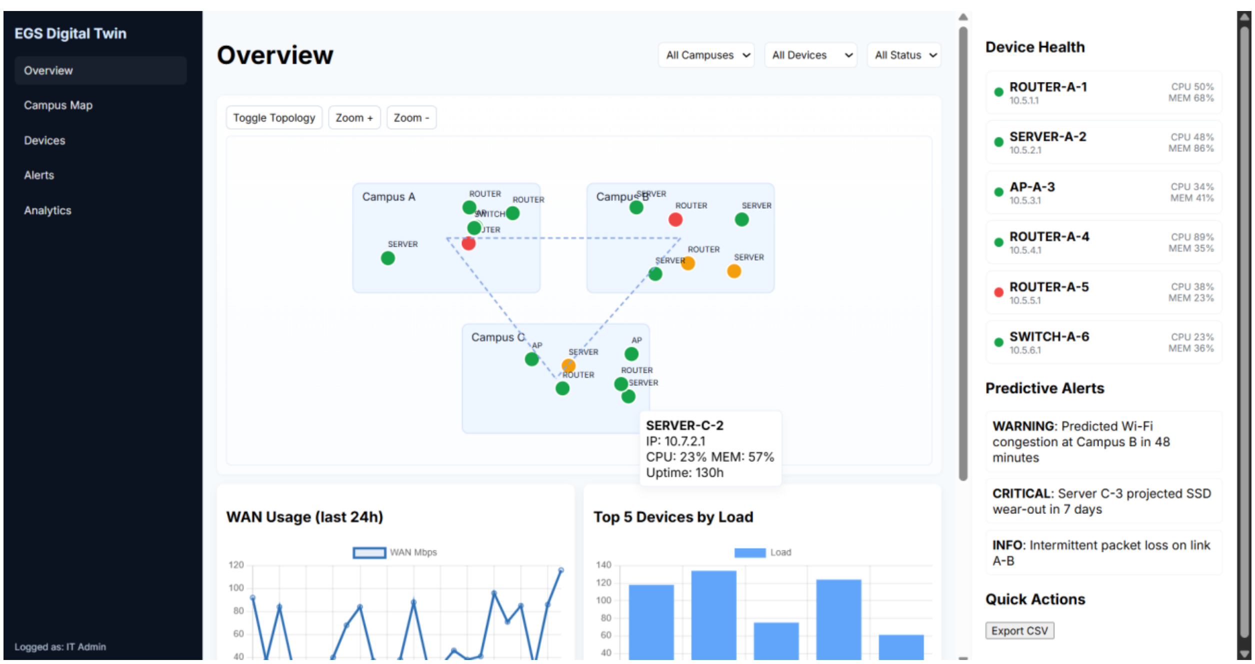

Fig. 6: Digital Twin Dashboard

The dashboard acts as the primary interface for this digital twin. It brings together device health information, live performance metrics, inter-campus connectivity, and analytics into one cohesive view. The aim is not only operational visibility but also strategic intelligence, enabling the institution to run a modern, scalable, and resilient digital infrastructure.

4.2.2. Mapping of Campuses, Floors, Rooms, and Devices

The system begins by representing each of the three campuses as an individual logical unit within the dashboard. Each campus contains its own internal layout, showing the servers, routers, switches, access points, and other networked assets. Devices are arranged in a way that reflects their placement in the actual environment, allowing administrators to trace how the network is structured. The digital twin provides a simplified but accurate view of floors and internal rooms, ensuring that the topology is intuitive while still reflecting the underlying architecture.

4.2.3. Visualization of Wired, Wireless, and Inter-Campus Links

Connections between devices are displayed visually to show how data flows within and across campuses. Wired links, Wi-Fi access routes, and WAN pathways are clearly drawn, allowing the IT team to understand the structure of the physical and logical network. A dotted link representation illustrates how the three campuses communicate with one another through their WAN connections. This helps highlight dependencies and shows how traffic is routed throughout the entire institution.

4.2.4. Status-Based Color Coding

To support rapid interpretation, devices in the dashboard are color-coded based on their operational state. Healthy systems appear in green, devices showing early warnings appear in yellow, and critical issues appear in red. This approach allows administrators to scan the dashboard and immediately identify areas requiring attention. When a device is clicked, detailed information appears, including IP address, CPU load, memory usage, uptime, and other operational parameters.

4.2.5. 2D and 3D Layer Representations

The platform supports both a simple 2D layout for general monitoring and an optional 3D representation for deeper analysis. The 2D view is ideal for day-to-day operations, providing a clear and uncluttered understanding of device placement and connectivity. The 3D layer is used to visualize stacked network layers, understand inter-floor routing, and trace how traffic moves between access points, switches, and core systems.

4.2.6. Redundancy Paths and Bottleneck Identification

The digital twin highlights redundant paths within the network, showing where backup routes exist for failover. At the same time, it identifies areas where only a single route is available or where load is accumulating. By analyzing device performance and traffic flow, the system can reveal bottlenecks that may affect bandwidth or latency during peak hours. This allows the school to take preventative action, either by upgrading hardware or adjusting configurations.

4.2.7. Prediction of Usage Patterns

The system continuously observes bandwidth demand and device load to predict future usage patterns. Based on current trends, the dashboard forecasts periods of high network activity, potential congestion zones, and devices likely to experience strain. AI-driven SD-WAN prediction enables proactive path selection, congestion avoidance, and automated routing decisions based on simulated and real-time network conditions. These predictive insights help the IT department plan capacity, manage peak traffic, and ensure the network remains stable during heavy academic usage, examinations, or schoolwide activities.

4.2.8. Device Health Monitoring

The right panel of the dashboard consolidates device health across all campuses in real time. It lists CPU and memory consumption for routers, servers, switches, and access points. This gives administrators a high-level snapshot of performance and helps them detect anomalies quickly. The ability to view the operational status of every major device enables faster troubleshooting and minimizes downtime.

4.2.9. Failure Prediction and Automated Alerts

The digital twin incorporates basic analytics to detect early signs of failure. If a server shows increasing SSD wear, if an access point is likely to experience congestion, or if a link begins to show intermittent packet loss, the system raises an alert. These warnings allow the IT team to intervene before service is affected. The alerts range from informational messages to time-sensitive warnings, supporting proactive maintenance instead of reactive repair.

4.2.10. Dashboard Illustration

The included dashboard image demonstrates how the digital twin integrates campus mapping, device status, inter-campus topology, performance graphs, and predictive alerts into a single interface. It shows how an administrator can view the state of all three campuses at a glance while still having the ability to explore individual devices and deeper diagnostic information. This visual serves as a representative example of how the proposed system would operate within the Edu-Group environment.

4.3. Cloud Infrastructure & Data Management

4.3.1. Overview

This section describes the cloud framework that supports the Digital Twin and Smart Management Dashboard. The cloud environment provides the computational backbone required for real time device monitoring, network topology management, predictive analytics, and historical data storage. It ensures that the digital representation of each campus remains continuously synchronized with the physical network while enabling multi-site scaling, high availability, and secure access to all operational data.

4.3.2. Cloud Deployment Model

The system uses a hybrid cloud deployment model that combines on premise edge processing with centralized cloud based computation. Edge devices handle initial filtering, preprocessing, and rapid local decision making. The cloud environment hosts the Digital Twin engine, analytics models, long term storage systems, and the services that supply data to the dashboard. This structure achieves low latency, resilience, and consistent performance across all campuses regardless of physical layout.

4.3.3. Data Storage Architecture

Data storage is organized into a tiered structure that separates high frequency telemetry, static metadata, and historical archives. Time series databases store bandwidth usage, device health parameters, and real time link performance. Object storage and relational databases maintain campus maps, floor plans, device inventories, and network configuration files. Older operational logs and maintenance records are transferred to long term archival storage so that they remain available for future analysis without affecting the performance of active systems.

4.3.4. APIs and Integration Framework

The cloud platform provides a set of secure REST and WebSocket APIs that enable communication between the physical network, the Digital Twin engine, and the management dashboard. Monitoring agents and discovery tools transmit updates to the cloud through authenticated endpoints. The dashboard retrieves processed information through lightweight APIs that supply data for 2D and 3D visualizations, device status updates, prediction results, and alert generation. This integration framework allows new systems or devices to be added without disrupting existing services.

4.3.5. Backup and Disaster Recovery

A structured backup and recovery strategy protects the continuity of the Digital Twin environment. Incremental backups of time series data are executed at scheduled intervals while full snapshots of configuration files and metadata databases are saved in geographically separated storage locations. Failover procedures allow the system to shift to standby cloud instances if the primary computational node becomes unavailable. Integrity checks, versioning controls, and rollback procedures ensure accuracy and reliability during restoration.

4.4. Cyber Security Architecture

A multi-layered Cyber Security Architecture is designed to protect the confidentiality, integrity, and availability of data across all three campuses. The security design follows a defense-in-depth approach, combining strong network controls, endpoint protections, identity and user governance, policy enforcement, and continuous threat monitoring. The architecture aligns with best practices from NIST CSF and ISO 27001 frameworks.

4.4.1. Network Security

Network security is implemented using a combination of VLAN segmentation, firewalls, SD-WAN encryption, and Access Control Lists (ACLs). Each campus is segmented into dedicated VLANs for Staff/Admin, Teachers, Students, IoT devices, Guest Wi-Fi, and critical Server/DMZ systems. This prevents sensitive internal services from mixing with high-risk networks, reducing the chances of lateral movement during an attack. Inter-VLAN communication is controlled by strict firewall policies and Layer 3 ACLs that only permit essential traffic flows.

Connectivity between campuses uses secure SD-WAN tunnels with built-in IPsec encryption, ensuring that data remains protected even when transmitted over public internet links. Perimeter firewalls add another layer of protection by enforcing intrusion prevention, web filtering, DNS security, and geo-blocking. Firewalls at each campus enforce intrusion prevention, web filtering, geo-blocking, and application control. Network Access Control (NAC) ensures that only compliant and registered devices can join staff and admin networks. This implementation minimizes attack surfaces, prevents rogue devices, and ensures secure campus-to-campus communication.

4.4.2. Endpoint & Device Security

Endpoint security is enforced through centralized antivirus/EDR solutions, automated patching, and device compliance controls. Staff and admin devices receive continuous protection against malware, ransomware, and unsafe applications. Updates and security patches are deployed centrally, ensuring devices remain protected without relying on users to manage updates manually. Servers hosting ERP systems and learning platforms are also hardened with limited-access privileges, secure configurations, and routine patch audits.

IoT and lab devices such as CCTV cameras, access control systems, projectors, biometric attendance devices, and Wi-Fi APs are isolated within a dedicated IoT VLAN with minimal access permissions. These devices can only communicate with their management servers or controllers, preventing them from being exploited as entry points into sensitive networks. Student devices connecting through Wi-Fi are automatically restricted to the Student network, ensuring that personal devices cannot access administrative systems.

4.4.3. Identity, Access Control & User Management

A centralized Identity and Access Management (IAM) system can be established based on Microsoft Azure AD. User authentication is handled via single sign-on (SSO), and Multi-Factor Authentication (MFA) is mandatory for privileged accounts such as IT administrators and system managers. Role-Based Access Control (RBAC) ensures users such as teachers, admin staff, students, IT teams can only access systems directly related to their roles. This prevents accidental misuse and reduces insider threat exposure.

User account lifecycle processes are automated. When students enroll, accounts are created with limited permissions for LMS access and Wi-Fi connectivity. When staff or students leave, accounts are automatically disabled and removed from groups, ensuring no abandoned accounts remain in the system. Password policies, device restrictions, and login monitoring further enhance identity security, maintaining a controlled and auditable access environment.

4.4.4. Security Policies, Governance & Compliance

Our proposed solution involves a security governance framework that ensures all systems, users, and devices operate within defined rules across the three campuses. Core policies include the Acceptable Use Policy, Password & Authentication Policy, Network Access Policy, BYOD Policy, and Data Protection Policy, which outline how staff, students, and administrators should use school systems such as Wi-Fi, ERP platforms, email, and IoT devices. They also define data handling requirements for sensitive information, enforce safe use of school laptops and smart boards, and restrict high-risk activities. Policies are reviewed annually or when new technologies such as IoT devices, ERP upgrades, or network expansions are introduced. This ensures the security framework continuously aligns with the school's operational needs.

Governance and compliance are maintained through consistent oversight by the IT Director and campus IT teams. Regular reviews of firewall rules, VLAN segmentation, SD-WAN policies, admin privileges, Azure AD accounts, and endpoint compliance ensure configurations remain secure and consistent across all campuses. A centralized monitoring dashboard tracks blocked threats, login anomalies, and device health, enabling rapid detection of policy violations. Azure AD enforces compliance automatically through MFA for administrators, password policies, conditional access, and immediate user deprovisioning when staff or students leave. This governance structure ensures that EduSchools remains secure, compliant, and well-managed without unnecessary complexity.

4.4.5. Threat Detection, Monitoring & Periodic Testing

A centralized monitoring system that collects logs from firewalls, SD-WAN appliances, servers, and endpoint security tools. These logs are analyzed for suspicious patterns such as repeated login failures, malware detections, unusual network traffic, or attempts to access restricted systems. Email filtering, DNS filtering, and web filtering automatically block phishing attempts, malicious websites, and harmful attachments before they reach end users.

Periodic security testing is conducted to maintain and validate the school's security posture. This includes routine vulnerability scans, Wi-Fi penetration audits, simulated phishing tests for staff awareness, and annual third-party penetration testing. Findings from these assessments are documented, prioritized, and remediated within defined timelines. This continuous improvement cycle ensures that EduSchools remain protected against evolving cyber threats.

4.4.6. Backup & Data Protection

To ensure the integrity and availability of critical school data, a robust backup and data protection strategy is implemented across its complete network. Regular automated backups are performed for servers, databases, and cloud-hosted applications, with both full and incremental backup schedules to minimize data loss. Advanced data protection measures, including encryption, access controls, and replication across multiple storage locations, safeguard sensitive information against accidental deletion, hardware failure, or cyber threats. These measures ensure that staff, students, and administrative operations can continue without disruption, even in the event of localized data corruption.

4.4.7. Security Testing

Security testing ensures that the EGS solution is robust against cyber threats and vulnerabilities, maintaining the confidentiality, integrity, and availability of organizational data.

1. **Objectives**
    - Verify the resilience of network devices (switches, access points, firewalls) against unauthorized access.
    - Ensure secure data storage and transmission across on-premises servers and cloud connections.
    - Validate compliance with organizational and regulatory security policies.
2. **Testing Methods**
    - **Vulnerability Scanning:** Use automated tools such as Nessus to detect weaknesses in network devices, servers, and storage systems.
    - **Penetration Testing:** Simulate real-world attacks on network endpoints, VPN gateways, and SD-WAN connections to uncover exploitable vulnerabilities.
    - **Firewall and Access Control Testing:** Verify Fortinet firewall rules, VLAN segmentation, and role-based access controls are effective.
    - **Configuration Review:** Check for proper hardening of servers, storage devices, and switches (like disabling unused ports, enforcing secure protocols).
    - **Patch and Update Verification:** Confirm all devices and software are updated to mitigate known security risks.
3. **Tools and Techniques**
    - Network monitoring tools (e.g., Wireshark, SolarWinds) for traffic analysis.
    - Security auditing tools (e.g., Nmap, Metasploit) for penetration testing.
    - Configuration management tools to ensure compliance with security standards.
4. **Reporting and Mitigation**
    - Document vulnerabilities, their risk level, and potential impact.
    - Implement mitigations such as firmware updates, rule modifications, or enhanced authentication mechanisms.
    - Perform retesting to confirm that vulnerabilities have been effectively resolved.

Outcome:
Comprehensive security testing ensures that the EGS solution is robust against cyber threats, secures sensitive data, and provides a resilient network infrastructure for enterprise operations.

4.5. Hardware & Software Infrastructure Design

4.5.1. Hardware Requirements

The proposed hardware seamlessly integrates with EGS's existing Aruba-based infrastructure while introducing advanced security and server capabilities through Fortinet appliances, HPE servers, and centralized NAS systems. Since IT staff are already familiar with Aruba technologies, transitioning to the upgraded network environment will be straightforward, supported by structured training focused on the new security, storage, and virtualization tools. This ensures the team can confidently operate and manage the enhanced multi-campus system while maintaining efficiency and reliability.

S.No.	Hardware Component	Recommended Model	Justification	Qty	Approx. Unit Cost (AED)	Total (AED)
1	Core Layer 3 Switch	Aruba 6300M 48-port (JL658A)	High-throughput backplane (370 Gbps), full Layer 3 routing, VSF stacking for redundancy, ideal for campus core aggregation	3	4,900	14,700
2	Access Layer Switch (PoE+)	Aruba 2930F 48G PoE+ (JL562A)	740W PoE+ budget supports multiple access points and IP phones, cost-effective with lifetime warranty	24	3,950	94,800
3	SD-WAN Edge Device	Fortinet FortiGate 60F (Secure SD-WAN)	Integrated SD-WAN with secure tunneling, ideal for branch sites up to 50 users with built-in threat protection	3	1,450	4,350
4	Next-Gen Firewall	Fortinet FortiGate 100F	10 Gbps firewall throughput with advanced threat protection, SSL inspection, and unified threat management for medium networks	3	2,350	7,050
5	Wi-Fi 6 Access Points	Aruba AP-515 Wi-Fi 6	Tri-radio Wi-Fi 6 with integrated spectrum analysis, high client density support (512+ devices), outdoor-rated IP67 enclosure	180	620	111,600
6	IoT	CCTV, NVR, Biometric Access, VoIP Phones, Smart Boards	Centralized IoT deployment for safety, automation & collaboration	3	9,000	27,000
7	Physical Servers	HPE ProLiant DL380 Gen11	Industry-standard 2U rack server with dual-socket scalability, enterprise-grade	4	2,150	8,600

			reliability, and extensive virtualization support			
8	NAS Storage System	Synology RS1221RP+ (8-bay)	Redundant PSU, 10GbE-ready, snapshot replication, and easy-to-manage DiskStation Manager for centralized storage	3	1,850	5,550
9	UPS Backup Systems	APC Smart-UPS 3000VA	2700W capacity with automatic voltage regulation, network management card compatible, 5-10 minute runtime for graceful shutdown	12	1,150	13,800
10	Cables and Physical Layer and	Cat6A Ethernet Cables, Fiber Optic Cables, RJ45 Patch Panels	Cat6A supports 10GBASE-T up to 100m, fiber for long-distance backbone links, structured cabling for professional installation	-	200,000	200,000
TOTAL						**AED 487,850**

4.5.2. Software Requirements

S.No.	Software / License	Qty / Users	Unit Cost (AED)	Approx. Cost (AED)
1	Microsoft 365 (Education Discount)	1,000	AED 200/year	AED 200,000/year
2	Windows Server License	3	AED 2600/year	AED 7,800/year
3	Windows Server CAL	1,000	AED 150/year	AED 150,000/year
4	SQL Server 2022 Standard	2	AED 6,000/year	AED 12,000/year
5	Azure VPN Gateway	1	AED 16,000/year	AED 16,000/year
6	Azure Virtual Desktop	100	AED 1,140 AED	AED 114,000/year
7	SKJ Unified Communication	1000	AED 150/year	AED 150,000/year
8	SKJ Digital Twin	30	AED 200/year	AED 6,000/year
TOTAL				**AED 655,800/year**

4.6. Unified Communication & Collaboration System

To create a seamless and efficient digital learning environment, Edu-Group Schools will implement a Unified Communication & Collaboration (UCC) framework. This system integrates messaging, meetings, learning platforms, and cloud-based collaboration into a single ecosystem. By unifying communication and educational tools, the framework enhances productivity, streamlines workflows, and ensures security and compliance across all campuses.

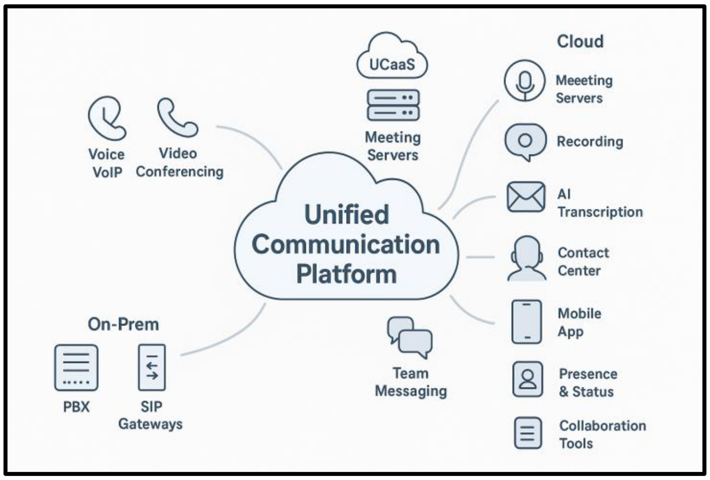

Fig.6: UCC Design

4.6.1. Key Functionalities

- **Seamless Unified Environment**
  The UCC system provides a consistent experience across all devices and platforms. Students, teachers, and staff can access communication, collaboration, and learning tools from a single interface, reducing friction and improving operational efficiency.

- **Single Sign-On (SSO) Experience**
  Centralized authentication simplifies user access while integrating Multi-Factor Authentication (MFA) to enhance security. This approach reduces password fatigue and strengthens protection of sensitive data.

- **Automated Course-to-Team Mapping**
  By linking Moodle course enrollments with Microsoft Teams channels, the system automatically connects students and teachers in relevant collaboration spaces. This automation eliminates manual setup and ensures timely and structured engagement.

- **Custom Dashboards & Reporting**

  Administrators benefit from centralized dashboards that track engagement, course completion, and collaboration metrics. Real-time reports provide actionable insights into learning outcomes and operational performance.

4.6.2. Platform Integration

Service	**Platform / Tool**	**Purpose**
Email & Calendar	Outlook / Exchange Online	Facilitates seamless email communication and scheduling across campuses.
Messaging & Meetings	Microsoft Teams	Supports real-time collaboration, chat, and video conferencing.
Learning & Assessments	Moodle LMS	Centralized learning management with automated gradebook synchronization.
Cloud Storage & File Collaboration	OneDrive & SharePoint	Secure document storage, version control, and team collaboration.

4.6.3. Extended Capabilities

The UCC framework extends beyond basic integration by offering advanced capabilities that enhance both learning and administrative efficiency. Centralized reporting dashboards consolidate data from all platforms, providing administrators with a holistic view of engagement, progress, and operational metrics. Personalized learning pathways enable the tracking of student performance, supporting customized learning journeys tailored to individual needs. The system ensures end-to-end integration, maintaining seamless interoperability between all tools while preserving data integrity. Additionally, robust security and compliance measures are implemented, including role-based access controls, single sign-on (SSO), multi-factor authentication (MFA), and comprehensive data protection policies, ensuring the platform meets institutional and regulatory standards.

4.6.4. Expected Outcomes

The implementation of this Unified Communication and Collaboration system will fundamentally transform Edu-Group Schools' ICT landscape into a highly connected, secure, and user-centric ecosystem. Students, teachers, and administrative staff will be able to collaborate seamlessly across campuses, access learning and operational resources in real time, and communicate effectively through a single integrated platform. By leveraging centralized dashboards, AI-driven analytics, and predictive monitoring, the system provides actionable insights that support data-driven decision-making, streamline workflows, and enhance operational efficiency. Ultimately, this transformation will not only improve teaching effectiveness and learning outcomes but also foster a more inclusive, responsive, and future-ready educational environment.

4.7. Existing vs. Proposed Solution

To illustrate the transformative impact of our ICT + Digital Twin system, the table below compares the existing Edu-Group Schools setup with the proposed enhanced solution. The upgraded system addresses challenges in network performance, communication, security, device management, cloud integration, and monitoring, while introducing predictive capabilities and seamless collaboration across campuses.

Category	Existing EGS Setup (Before)	Our Enhanced ICT Solution (After)
Network Design	Mixed legacy and new hardware, each campus isolated, inconsistent performance	Unified SD-WAN multi-campus fabric with guaranteed performance and a centralized command center
Communication Tools	Fragmented: Teams, ERP messages, WhatsApp groups, email; nothing synced	Unified Communication Platform integrating chat, voice, video, alerts, LMS, and ERP
Wi-Fi Performance	Newer Wi-Fi 6 APs only in some buildings, older APs causing dead zones	AI-optimized Wi-Fi with RF heatmaps, automatic AP tuning, and real-time coverage monitoring
Security	Basic firewall and endpoint protection; no segmentation; high risk with BYOD	Zero-Trust segmentation, dynamic ACLs, MFA, IDS/IPS, and encrypted traffic across campuses
Device & IoT Management	No centralized mapping; cameras, boards, tablets all added manually	Full IoT asset management with automated onboarding and a digital twin of all smart devices
Cloud Usage	Limited to OneDrive and Teams; on-prem ERP servers	Hybrid-cloud model with scalable hosting for LMS, ERP, and backups
Monitoring & Troubleshooting	Mostly manual; no predictive analytics; issues found after failures	AI predictive maintenance, fault forecasting, automated alerts, and proactive remediation
Scalability	Bandwidth bottlenecks as student numbers grow; inconsistent topology	Elastic SD-WAN scaling, cloud-ready architecture, modular upgrades, and future-proof design
Simulation Ability	None; changes applied blindly, high risk of downtime	Full network digital twin simulation for testing configurations before deployment
Parent-Teacher-Student Collaboration	Siloed platforms; inconsistent information flow	One connected platform enabling seamless messages, announcements, and class updates

4.8. Competitive Benchmark

While several commercial solutions such as Cisco Meraki, Huawei CloudCampus, HPE Aruba, Juniper Mist AI, and Azure Digital Twin offer varying levels of network management and digital infrastructure support, Edu-Group Schools' ICT + Digital Twin solution stands out due to its comprehensive, end-to-end approach. Many platforms are limited to specific domains, such as Wi-Fi optimization, building IoT, or closed ecosystems. In contrast, our solution provides full Digital Twin simulation across the entire network, enabling realistic pre-deployment testing and predictive scenario planning.

Network-wide AI-driven predictive maintenance allows administrators to address potential issues proactively rather than focusing only on Wi-Fi or isolated systems. The solution also provides hybrid-cloud flexibility with AWS, Azure, and private cloud deployments while maintaining complete visibility across LAN, WAN, and Wi-Fi networks. Security is enhanced through fully automated zero-trust segmentation and dynamic access control lists, along with a multi-campus centralized control center that unifies command and monitoring across all locations.

Additionally, the platform offers user experience analytics across applications and devices, AI-optimized Wi-Fi coverage integrated with the Digital Twin, and cost-efficient, vendor-neutral deployment options. Risk-free pre-deployment testing allows administrators to simulate network changes and predict outcomes before implementation, a capability not available in most conventional solutions. Together, these features make the ICT + Digital Twin system a unique, future-ready network solution that combines operational intelligence, robust security, and holistic management for educational institutions.

5. Management and Governance Plan

5.1. Timeline and Phase-wise Project Plan

This phased implementation plan outlines the structured progression of the smart-campus upgrade, detailing each stage from initial planning through deployment, testing, and final handover. By breaking the project into clear, time-bound phases with defined activities and deliverables, the team ensures an organized workflow, smooth coordination, and a predictable path toward a fully optimized and future-ready network ecosystem.

S.No.	Phase	Duration	Activities / Plan	Deliverables / Outcomes
1.	Project Initiation & Planning	Week 1	Requirements gathering, stakeholder interviews, existing infrastructure survey, gap analysis	Requirement document, survey report, project charter
2.	Network Design & Architecture	Week 2–3	Design VLANs, routing, IP schema, Wi-Fi coverage, server roles, cloud architecture	Network architecture document, topology diagrams, design approvals

3.	Hardware Procurement & Setup	Week 4–10	Procure switches, routers, APs, servers; vendor orders; staging setup; device imaging	Hardware inventory, configured baseline environment ready for deployment
4.	Network Configuration & Implementation	Week 11–12	Configure VLANs, routing, QoS, firewall policies; deploy APs; validate performance	Fully deployed LAN/WLAN network, testing logs
5.	Digital Twin Integration	Week 12-13	Deploy Digital Twin engine, connect monitoring agents, onboard devices, configure dashboards	Real-time monitoring dashboards, predictive maintenance enabled
6.	Unified Communication Setup	Week 14	Implement VoIP, SIP, collaboration suite, video conferencing integration	Functional unified communication system
7.	Cybersecurity & Testing	Week 14–15	Endpoint security deployment, network segmentation review, vulnerability scanning, load testing	Security posture report, test results, compliance validation
8.	Training & Handover	Week 16-17	Train staff on operations, dashboard usage, SOPs; hand over documentation	Trained IT team, operations manuals, final report
9.	Post-Deployment Support	Week 17–18	Monitor network, fine-tune configurations, resolve early-stage issues	Optimization report, issue resolution log
10.	Contingency/Buffer Period	Week 19-20	Handle delays in procurement, testing, training, or unforeseen issues	Ensure project milestones are met on schedule

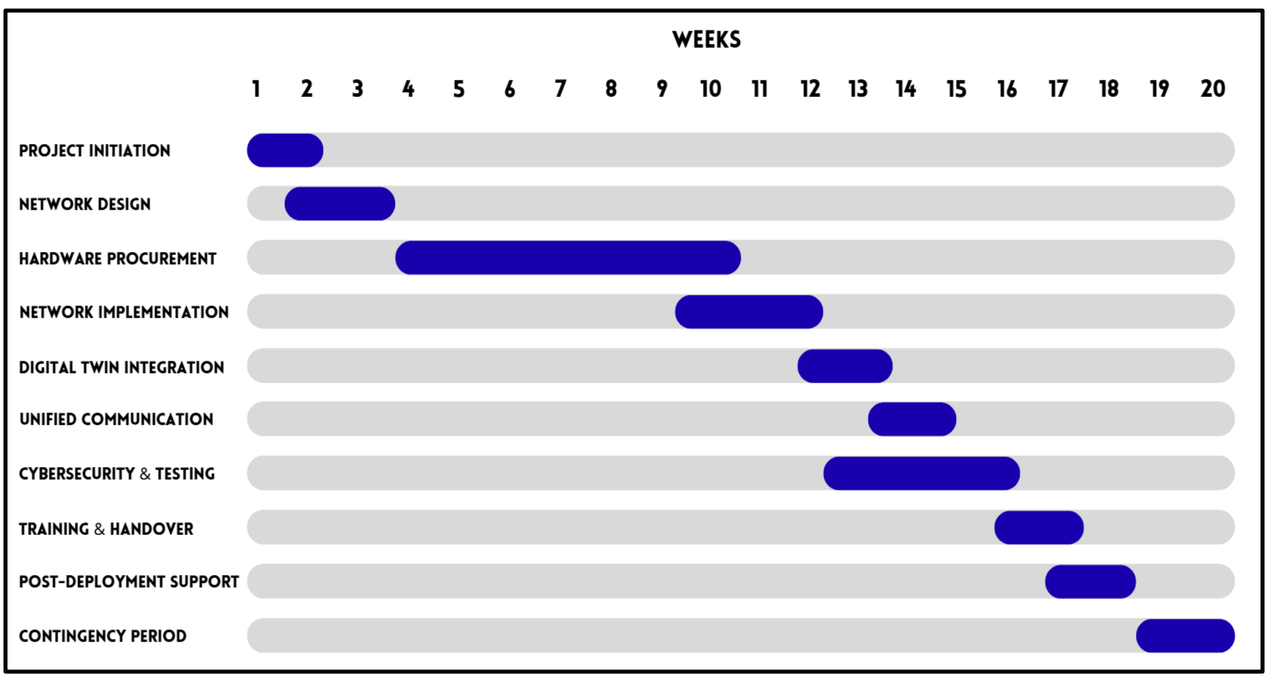

Fig. 7: Timeline Gantt Chart

5.2. Risk Assessment & Mitigation

A successful smart-campus transformation depends on anticipating challenges before they disrupt progress. This Risk Assessment and Mitigation Plan outlines the key technical, operational, and compliance risks that may arise during the project, along with their likelihood, impact, and targeted strategies to minimize disruption. By identifying these risks early and preparing structured responses, the team ensures smoother execution, stronger system reliability, and a resilient overall project rollout.

Risk Area	Description of Risk	Likelihood	Impact	Overall Risk Level	Mitigation Strategy
Hardware Procurement Delays	Delivery of servers, switches, firewalls, or access points may be delayed, affecting installation timelines.	Medium	High	High	Pre-order critical items early, confirm vendor lead times, keep backup suppliers, and prioritize components for essential network zones.
Network Configuration Errors	Misconfigured VLANs, routing rules, APs, or firewall policies may cause outages.	Medium	High	High	Use standardized configuration templates, peer review configurations, conduct lab testing before deployment, maintain rollback versions.
Cybersecurity Vulnerabilities	Misconfigurations or outdated firmware may expose the school to cyberattacks or data breaches.	High	High	Very High	Enforce regular patching, conduct vulnerability scans, adopt Zero-Trust segmentation, enable MFA, monitor logs through SIEM tools.
Digital Twin Data Inaccuracy	Incorrect or inconsistent real-time data could lead to faulty modelling, wrong predictions, or misleading dashboards.	Medium	Medium	Medium	Validate sensors, implement calibration schedules, build data validation layers, and cross-check readings with historical baselines.
Integration Failures	Problems integrating IoT systems, energy sensors, attendance systems, and network monitoring tools with the Digital Twin.	Medium	High	High	Define clear API standards, test each integration in a sandbox, maintain documentation, schedule iterative integration cycles.

Wireless Coverage Gaps	Poor WiFi performance in certain blocks due to interference or poor AP placement.	Medium	Medium	Medium	Conduct a site survey (heatmap), adjust channel planning, place APs strategically, and introduce additional APs if needed.
User Resistance to New Systems	Staff may struggle adapting to new dashboards, smart systems, or workflow automations.	Low	Medium	Low–Medium	Provide training workshops, create short user manuals, set up a helpdesk support phase, and gather early feedback.
Power or Cooling Issues	Server room overheating or power fluctuations can cause downtime.	Low	High	Medium	Install UPS capacity, monitor temperature/humidity through sensors, ensure proper RACK airflow, create an escalation SOP.
Vendor Support Delays	Third-party vendors may fail to respond quickly during troubleshooting.	Medium	Medium	Medium	Establish SLAs, maintain 24/7 contact channels, document escalation paths, and train in-house IT staff for basic maintenance.
Data Privacy Non-Compliance	Violation of student/staff data regulation during network logging or digital twin modelling.	Low	High	Medium	Follow data protection policy, anonymize sensitive data, enforce access control, and conduct compliance reviews.

6. Financial Strategy

The proposed ICT solution is supported by a flexible business model designed to meet the strategic and financial requirements of Edu-Group Schools. To accommodate institutions with varying budget structures and procurement preferences, SKJ Solutions offers two distinct deployment and payment models:

- Asset Ownership Model
- NaaS Subscription Model

Both models provide the same high-quality ICT infrastructure, cybersecurity framework, cloud integration, and Digital Twin capabilities. However, they differ in the approach to ownership, cost distribution, and long-term financial impact. This dual-model strategy ensures that Edu-Group Schools can select the structure that aligns best with its financial policy, budget cycle, and operational priorities.

6.1. Business Model

SKJ Solutions provides a deployable model through a structured combination of one-time, recurring, and expansion-based needs. These streams differ depending on whether the school chooses the Ownership model or the NaaS model.

6.1.1. Model 1: Asset Ownership-Based Deployment

The Asset Ownership Model enables EGS to fully own the ICT infrastructure while benefiting from SKJ Solutions' deployment expertise, cybersecurity framework, cloud integration, and Digital Twin capabilities.

1. **One-Time Investment (CapEx)**

This stream represents the upfront investment required for procuring, deploying, and configuring the ICT infrastructure. It covers all physical and software assets needed for a fully operational network.

- Hardware procurement: network switches, firewalls, servers, access points
- Structured cabling & fibre backbone
- SD-WAN deployment
- Digital Twin setup
- ICT installation & configuration
- Initial training and documentation

2. **Recurring Expenses (OpEx)**

Recurring expenses are generated from ongoing operational services, software subscriptions, cloud hosting, and maintenance. This ensures the network remains secure, efficient, and up-to-date while minimizing downtime and manual intervention.

- Cloud hosting for Virtual HQ
- Azure AD & M365 subscription management
- Digital Twin SaaS subscription
- Cybersecurity & endpoint protection
- 24/7 monitoring & support
- Backup and disaster recovery

3. **Add-On / Expansion Expenses**

This stream arises from additional services, upgrades, and extensions to meet evolving institutional needs. It allows the school to scale the network, enhance learning environments, and adopt technologies over time.

- Additional APs, cameras, IoT devices
- New campus integration
- Smart classroom systems
- AI analytics & upgrades
- IT staff workshops

This model benefits schools seeking full ownership and control of their ICT infrastructure, enabling long-term asset value and flexible expansion. It is best suited for institutions with stable capital budgets and in-house IT capability.

6.1.2. Model 2: Network-as-a-Service (NaaS) Model

In the NaaS model, SKJ Solutions provides the entire network infrastructure as a subscription service, eliminating the need for upfront hardware investment. The institution pays a monthly or yearly fee based solely on usage and scale, turning the network into a fully OpEx-based service.

Under NaaS, SKJ Solutions retains ownership of:

- Hardware (switches, APs, firewalls, edge devices)
- Software licenses
- Cloud resources
- Monitoring and automation tools

The school receives:

- A fully managed, end-to-end network
- Automatic upgrades and replacements
- Predictable monthly operating costs
- Scalability without major reinvestment
- SLA-backed performance and uptime guarantees

This model benefits institutions by reducing upfront costs, offering flexible budgeting, eliminating maintenance responsibilities, and enabling effortless scalability as their network needs expansion.

6.2. Pricing Model

SKJ Solutions offers two distinct pricing models designed to provide Edu-Group Schools with flexibility in financing their complete ICT infrastructure transformation. Each model delivers the same comprehensive solution but differs in payment structure and ownership terms.

6.2.1. Model 1: Asset Ownership Model

This model follows a conventional approach where the institution purchases all infrastructure components outright and pays for ongoing operational services separately.

Capital Expenditure (CapEx) - One-Time Payment

Component Category	**Cost (AED)**
Hardware Subtotal	487,450
Professional Installation & Deployment	85,000
Digital Twin Initial Setup	25,000
Staff Training & Handover	12,000
Total CapEx Investment	**609,450**

Operational Expenditure (OpEx) - Annual Recurring

Service Category	Annual Cost (AED)
Software Licenses (Windows Server, M365, Azure AD, Endpoint Security)	655,800
Cloud Infrastructure Hosting (Virtual HQ, Applications, Storage)	18,000
Cybersecurity Services (Threat Intelligence, Monitoring, Updates)	18,000
Technical Support & Managed Services (24/7 NOC)	36,000
Backup & Disaster Recovery Services	12,000
Total Annual OpEx	**739,800**

Cost Summary

- **Initial Investment (Year 0):** AED 609,450
- **Year 1 Total Cost:** AED 1,349,250 (CapEx + First Year OpEx)
- **Annual Cost (Years 2+):** AED 739,800
- **5-Year Total Cost:** AED 4,308,450

6.2.2. Model 2: NaaS Subscription Model

This model provides the complete ICT solution through a comprehensive monthly subscription that includes hardware leasing, software licensing, cloud services, maintenance, and full support while eliminating large upfront capital requirements.

NaaS Pricing Structure

Component	Monthly Cost (AED)
Setup & Deployment Fee (One-Time)	120,000
Monthly Subscription Fee (All-Inclusive)	85,000

Contract Terms:

- **Minimum Contract Period:** 60 months (5 years)
- **Setup & Deployment Fee (One-Time):** AED 120,000
- **Annual Subscription Cost:** AED 1,020,000
- **Total 5-Year Contract Value:** AED AED 5,220,000

Cost Summary

- **Initial Payment (Setup):** AED 120,000
- **Monthly Payment:** AED 128,500
- **Year 1 Total Cost:** AED 1,140,000
- **5-Year Total Cost:** AED 5,220,000

6.3. Return on Investment (ROI)

The proposed ICT transformation offers substantial financial and operational returns for Edu-Group Schools. The ROI is realized through reductions in downtime, improved security resilience, and efficiency gained through automation and cloud integration.

1. **Reduced Downtime**
   The implementation of SD-WAN, improved Wi-Fi coverage, and centralized monitoring reduces annual downtime from approximately 50–70 hours to less than 5 hours. This improvement translates to an estimated AED 25,000–50,000 in annual productivity savings per campus.

2. **Cyber Risk Reduction**
   The multi-layered security architecture which includes MFA, endpoint protection, network segmentation, and continuous threat monitoring, reduces ransomware and malware exposure by 80–90%. Avoided incident costs are valued at approximately AED 100,000 or more per major event, along with reduced IT interventions.

3. **Predictive Monitoring & Automation**
   The Digital Twin platform enables predictive maintenance, automated alerts, and faster troubleshooting by 40–60%. As a result, IT labor, manual diagnostics, and reactive repairs are significantly minimized, generating annual savings of AED 100,000–150,000.

4. **Cloud Efficiency & Operational Optimization**
   Replacing physical HQ servers with scalable cloud hosting reduces energy consumption, cooling requirements, and hardware maintenance. This yields an additional AED 100,000–120,000 in annual savings.

Combined, the solution provides an estimated annual financial benefit of AED 250,000–350,000, resulting in a projected payback period of 2–3 years under the CapEx/OpEx model. Under the NaaS subscription model, cost neutrality is achieved immediately due to the absence of upfront capital expenditure.

Total Annual Financial Benefit:

➔ AED 250,000 – 350,000

Estimated Payback Period:

➔ 2–3 years under CapEx + OpEx

➔ Immediate cost neutrality under NaaS due to zero upfront investment

7. Future Improvements

The current EGS solution has been designed to be cost-friendly and easily adaptable, making it practical for immediate deployment and integration. Its simplicity ensures minimal setup complexity while providing reliable performance for the intended use case.

Looking ahead, if EGS seeks a more advanced version of the system, several enhancements can be considered:

1. **Enhanced Automation and AI Integration**
   Incorporating machine learning algorithms or predictive analytics could allow the system to anticipate operational needs, optimize performance, and improve decision-making efficiency.

2. **IoT Sensor Expansion**
   Integrating additional IoT sensors would enable real-time monitoring of environmental conditions, operational metrics, and user interactions, providing richer data for improved decision-making.

3. **Self-Healing Network Capabilities**
   Future iterations can include self-healing network features that automatically detect faults or connectivity issues and reconfigure themselves to maintain uninterrupted operation.

4. **User Experience Analytics**
   Advanced analytics can track system usage patterns and user interactions, offering insights to refine workflows, enhance usability, and ensure optimal performance from an end-user perspective.

5. **Advanced Security Features**
   Implementing multi-layered cybersecurity protocols, such as zero-trust, real-time intrusion detection, and automated threat response mechanisms could strengthen the system against evolving digital threats.

6. **Energy Efficiency and Sustainability**
   Future upgrades could focus on reducing energy consumption through smart power management and integrating renewable energy support, enhancing both sustainability and operational cost-efficiency.

Overall, the current solution provides a robust, economical, and easy-to-deploy foundation. With these potential improvements, EGS could achieve higher performance, greater intelligence, enhanced adaptability, and improved user satisfaction, ensuring long-term value and future readiness.

8. Conclusion

The implementation of a comprehensive ICT solution for Edu-Group Schools represents a significant leap toward modernizing its digital infrastructure across multiple campuses. By leveraging SD-WAN-enabled network architecture, advanced cybersecurity frameworks, Digital Twin-powered monitoring, and unified communication platforms, SKJ Solutions has designed an ecosystem that is secure, scalable, and intelligent. The proposed solution ensures high-performance connectivity, seamless inter-campus collaboration, and real-time visibility into network operations while maintaining strict compliance with cybersecurity and data protection standards.

Through careful network segmentation, predictive monitoring, and integration of cloud-based services with on-premises infrastructure, the project addresses both current operational needs and future growth. The inclusion of robust hardware, software, and unified communication systems ensures that classrooms, administrative offices, and IoT-enabled environments operate efficiently and reliably. Furthermore, the Digital Twin platform empowers administrators with actionable insights, proactive maintenance, and data-driven decision-making capabilities, enhancing overall operational intelligence.

The phase-wise implementation plan, risk assessment framework, and governance policies provide a structured and resilient approach to deployment, minimizing potential disruptions and ensuring continuity of educational operations. Ultimately, this project positions Edu-Group Schools at the forefront of digital educational infrastructure, fostering a future-ready, secure, and intelligent learning environment that enhances teaching outcomes, operational efficiency, and long-term sustainability.

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10. Appendix

10.1. Appendix A: Supporting Links

10.1.1. Solution Pitch Link:

https://www.canva.com/design/DAG44PUv8q4/e3spIYwPfX24k-Hl5zHa6A/view?

10.1.2. Dashboard Link:

https://ak6325.github.io/ECTE-364-PROJECT-DASHBOARD/

10.2. Appendix B: Group Checklist

School of Engineering

University of Wollongong in Dubai

Group Members

Std Member	Name	Email	Signature
8183302	Ananya Lahari Vadlamani	alv737@uowmail.edu.au	
8272803	Vanshika Hemendra Kumar Choube	vhkc621@uowmail.edu.au	

Weekly Tasks (to be completed week by week)

Week	Task
Week 1	Group Formation – First Formal Meeting
Week 2	Ideation & Research
Week 3	Concept Finalization & Requirements Gathering
Week 4	Research & Background Study
Week 5	Design Phase
Week 6	Implementation Planning
Week 7	Draft Report Due
Week 8	Prototype Development & Testing
Week 9	Solution Presentation
Week 10	Final Report & Feedback Integration
Week 11	Final Report Due

Group Members

Std Member	Name	Signature	Contribution
8183302	Ananya Lahari Vadlamani		50 %
8272803	Vanshika Hemendra Kumar Choube		50 %
			100 %

Actual Tasks Completed by Members

Name	Tasks
Ananya Lahari Vadlamani	● Company Background ● ICT Solution - Network Architecture & Design ● ICT Solution - Cloud Infrastructure & Management ● ICT Solution - Cyber Security Architecture ● ICT Solution - Hardware and Software Infrastructure Design ● ICT Solution - Unified Communication & Collaboration ● Business Model - Business Model ● Business Model - Pricing Model ● Business Model - Return of Investment (ROI) ● Future Improvements
Vanshika Choube	● Executive Summary ● Introduction ● Company Background ● ICT Solution - Digital Twin Dashboard ● ICT Solution - Cloud Infrastructure & Management ● ICT Solution - Unified Communication & Collaboration ● ICT Solution - Existing vs. Proposed Solution ● ICT Solution - Competitive Benchmark ● Management and Governance Plan - Timeline and Phase-wise Project Plan ● Management and Governance Plan - Risk Assessment & Mitigation ● Future Improvements ● Conclusion

Comments: